

Chapter 6: Preparation of Small-Molecule Libraries

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1 Introduction

Small molecules that bind receptors are called ligands. Although the terms *ligand* and *drug* are sometimes used interchangeably, it is important to recognise that while all drugs are ligands, not all ligands qualify as drugs. Ligands are typically represented initially as 2D atomic connectivity graphs. These representations are converted into 3D structures and subjected to energy minimisation before molecular docking simulations. When screening large chemical libraries, often comprising millions or even billions of compounds, an initial filtering stage is commonly applied to enrich for molecules with drug-like properties. This process typically involves evaluating physicochemical descriptors such as molecular weight, lipophilicity (log P), the number of rotatable bonds, and the counts of hydrogen bond donors and acceptors. One of the most widely applied guidelines for such filtering is Lipinski's Rule of Five, originally proposed by Christopher Lipinski[1]. It should be noted that the Rule of Five primarily applies to orally administered drugs that permeate biological membranes through passive diffusion rather than active or receptor-mediated transport mechanisms. Nevertheless, these filters remain valuable for excluding compounds that are unlikely to exhibit suitable pharmacokinetic properties. In addition to physicochemical filtering, the identification and removal of pan-assay interference compounds (PAINS)[2] constitute an important step in early screening workflows. PAINS are chemical structures that frequently appear as active across multiple assay systems and are therefore often associated with assay interference or false-positive activity.

In most docking campaigns, the computational protocol is only as reliable as the quality of the input receptor structure, and equally critically, the representation and preparation of small molecules. Inadequate ligand preparation, such as incorrect protonation, missing tautomers, unrealistic conformations, or inconsistent valence and stereochemistry, can lead to misleading enrichments, even if the underlying search and scoring algorithms are otherwise robust.[3–5]

This chapter aims to provide a systematic and in-depth overview of methods and software for preparing and handling small molecules for molecular docking. The focus is on both theoretical concepts (e.g., protonation equilibria, tautomerism, conformational sampling, charge models) and practical aspects (software packages, file formats, and databases). Emphasis is placed on reproducible workflows, verification of structural correctness, and the use of curated ligand datasets and benchmarks.

Ligand preparation, therefore, aims to:

1. Standardise connectivity and valence to ensure correct bond orders, aromaticity, and formal charges consistent with chemical intuition and experimental pK_a /redox state.
2. Generate relevant protonation and tautomeric states to enumerate or select those states likely populated at the target pH and environment.
3. Resolve stereochemistry and isomerism to appropriately define chiral centres and E/Z double-bond stereochemistry, enumerating alternatives if necessary.
4. Generate low-energy 3D conformations by sampling accessible conformational space so that docking can identify plausible binding modes.
5. Assign force field parameters and partial charges to provide optimal parameters required by the docking engine’s scoring function.
6. Filter and quality-control to remove impossible or highly strained structures, salts, and reactive or toxic chemotypes.

Extensive benchmarking has shown that systematic attention to these steps can markedly improve virtual screening enrichment factors, sometimes by a factor of 2 or more[3, 6].

1.1 Typical workflow

A typical ligand preparation workflow for docking can be outlined as:

1. Input 1D/2D structures (SMILES or SDF) and perform basic corrections, such as valence checks, and standardisation.
2. Delete salts and solvents atoms.
3. Predict and enumerate relevant protonation states and tautomers at specified pH (e.g., 7.4 ± 1).
4. Enumerate undefined stereocenters and E/Z isomers, within user-defined limits.
5. Generate 3D coordinates for each state and perform conformational sampling.
6. Assign partial charges and any special atom types needed by the docking program.
7. Perform restrained energy minimisation using a force field suitable for small molecules.
8. Export to the docking engine’s preferred file format (e.g., MOL2, PDBQT, or SDF).
9. Verify geometry and chemistry with automated and manual checks.

Software such as Open Babel[7], RDKit (with ETKDG-based embedding)[8], OMEGA[9], and Gypsum-DL[10] implement many of these steps in open-source form, while commercial suites (Schrödinger, OpenEye, MOE, Discovery Studio) provide additional proprietary functionality.

2 Representation and Basic Processing of Small Molecules

2.1 1D, 2D, and 3D representations

Small molecules may be initially represented as:

- 1D line notations: e.g., SMILES[11], InChI[12]. These encode molecular graphs as strings, convenient for storage in large databases and substructure/similarity search.
- 2D connection tables: e.g., MDL MOL/SDF, which store atom types, bond orders, and 2D coordinates. Molecular formats such as SDFs allow the storage of molecular properties.
- 3D coordinate files: e.g., MOL2, PDB, SDF or PDBQT, which include 3D coordinates and partial charge information suitable for docking. The amount of information needed depends on the software one might choose for docking.

Docking requires 3D geometries; however, the ligand databases, such as, ChEMBL[13], ZINC[14], and PubChem[15] often provide structural information in 1D SMILES or 2D SDF formats, necessitating the generation of 3D coordinates and conformers.

2.2 Sanitisation

Most cheminformatics toolkits implement a sanitisation step that verifies chemical correctness. These include valence checks to ensure that valence counts of atoms conform to allowed patterns (e.g., tetravalent carbon, trivalent nitrogen in amines). Furthermore, valence checks also aids in correcting missing bond orders from connectivity. Since most drugs are aromatic in nature, a special emphasis on aromaticity is crucial and by applying models such as, Huckel-like rules or pre-tabulated aromatic atom types, are helpful to label aromatic rings. Lastly, the partial assignment to ensure the total charge is consistent with valence and, where possible, with expected pK_a .

In RDKit[16], for example, sanitisation is performed via `Chem.SanitizeMol`, which checks valence, aromaticity, conjugation, and ring information, raising errors or warnings when invalid structures are encountered. Open Babel[7] provides analogous functionality, including detection of impossible valence and inconsistent aromaticity. Sanitization should be done *before* any optimization or docking-related processing.

2.3 Standardization

Standardisation aims to convert diverse input representations into a consistent internal representation. Common steps include:

- Deleting salts to remove counterions (e.g., chloride, sodium) and solvents (e.g., water, DMSO) using fragment-based rules.
- Normalisation of functional groups to convert alternative representations (e.g., nitro as $\text{N}^+(\text{=O})[\text{O}^-]$) into a canonical pattern.
- Neutralisation to adjust charges on strong acids/bases to a standard reference pH or to a neutral form.
- Stereochemistry cleanup to remove or set undefined chiral centres when the input stereo is ambiguous.

Standardisation libraries, such as e.g., RDKit’s `rdMolStandardize` and commercial standardisation tools, encapsulate rules derived from medicinal chemistry practice.

3 Protonation States, Tautomerism, and Stereochemistry

3.1 Protonation equilibria and the Henderson–Hasselbalch equation

The protonation state of ionizable groups, such as carboxylic acids, and amines, is governed by acid-base equilibria. For a monoprotic acid HA,



The Henderson–Hasselbalch equation gives the ratio of deprotonated to protonated species at equilibrium:

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]} \quad (2)$$

Rearranging,

$$\frac{[\text{A}^-]}{[\text{HA}]} = 10^{\text{pH} - \text{p}K_a} \quad (3)$$

For a given pH and known pK_a , this allows the computation of the fractional populations of each protonation microstate. For multiple ionizable centres, microstate populations can be computed from a micro- K_a or micro- pK_a model; in practice, software such as Epik[17], ChemAxon pK_a predictors, or Dimorphite-DL[18] (used within Gypsum-DL[10]) implement empirical or semi-empirical models to enumerate and score protonation microstates[3].

In docking, it is rarely sufficient to consider a single protonation state; instead, all states within a reasonable free energy window (often a few kcal/mol at the working pH) should be generated. The Boltzmann weight of each state i is

$$p_i = \frac{e^{-\beta G_i}}{\sum_j e^{-\beta G_j}}, \quad \beta = \frac{1}{k_B T} \quad (4)$$

where G_i is the free energy of microstate i , and k_B is the Boltzmann constant. At a typical screening pH, states whose populations are below 1–5% may be neglected, however, borderline cases, such as histidines, and anilines, can strongly influence binding modes.

3.2 Tautomerism

Many heterocycles and functional groups exhibit tautomerism, such as keto-enol, amide-imidic acid, and lactam-lactim. Tautomers often differ in hydrogen placement and formal charges, with potentially large effects on hydrogen bonding and electrostatics in the binding site. Automatic tautomer enumeration proceeds by defining substructure-based transformation rules (SMARTS patterns) that map between tautomeric forms.

In practice:

- Enumerators generate all possible tautomers under a given set of rules.
- Scoring functions or empirical rules estimate relative stability (e.g., favouring amide over imidic acid forms).
- A subset of low-energy tautomers is selected for docking, sometimes weighted by estimated populations.

Tools such as Gypsum-DL explicitly enumerate tautomers (and other isomeric forms) and produce 3D structures for each selected form[10]. Commercial programs (e.g., LigPrep, Epik, and MOE) implement proprietary tautomer generators and scoring schemes[3].

3.3 Stereochemistry and E/Z isomerism

Stereochemical ambiguities are common in large virtual libraries. The chiral centres may be unspecified (e.g., input from generic SMILES), while the E/Z configuration of alkenes may be unknown or partially constrained. Atropisomerism and conformational chirality are usually not explicitly encoded. A general strategy is to keep experimentally known stereochemistry fixed from curated databases like ChEMBL or PDB ligands. This is followed by enumerating all reasonable stereoisomers for undefined centres; however, cap the total number of isomers per molecule to control combinatorial explosion. And for flexible or unassigned stereocenters, consider whether stereochemistry is relevant to the docking hypothesis. Gypsum-DL[10], for example, can enumerate R/S stereocenters, E/Z isomers, and ring conformers, within user-defined limits, for each input molecule.

4 3D Coordinate Generation and Conformational Sampling

The goal of conformer generation is to produce an ensemble of low-energy 3D geometries that approximate the accessible conformational space of a molecule in solution, while respecting chirality and other stereochemical constraints. Broadly, two classes of methods are used:

Systematic or rule-based sampling : Rotatable bonds are identified, and torsions are sampled from libraries of preferred angles; combinatorial explosion is controlled by pruning and energy filters.

Stochastic sampling : Monte Carlo or distance geometry-based methods sample conformational space randomly or quasi-randomly, often combined with local minimisation.

4.1 OMEGA

OMEGA (OpenEye)[9] is a widely used systematic conformer generator for small molecules and has become a *de facto* standard in many docking workflows. Its algorithm operates in three phases, first is a fragment-based initial structure assembly. In this phase, molecules are decomposed into frag-

ments at rotatable bonds; fragment libraries derived from crystallographic data provide 3D templates. The next phase is exhaustive torsion enumeration in which rotatable bonds are sampled at discrete torsion angles from a knowledge-based torsion library. This generates a potentially large raw conformer set. And the third phase is pruning by geometry and energy in which conformers that are too similar, using RMSD threshold (**Equation 5**), or too high in strain energy are discarded to yield a diverse low-energy ensemble. OMEGA[9] has been extensively validated against high-quality crystal structures from the PDB and Cambridge Structural Database[19], demonstrating high success rates in reproducing experimental ligand conformations within low RMSD thresholds.

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\| \mathbf{r}_i^{(\text{ref})} - \mathbf{r}_i^{(\text{fit})} \right\|^2} \quad (5)$$

where N is the number of matched atoms, and $\mathbf{r}_i^{(\text{ref})}$ and $\mathbf{r}_i^{(\text{fit})}$ are their coordinates after optimal superposition.

4.2 Distance geometry

Distance geometry (DG) methods[20] (See **chapter 2** for in-depth discussion) construct 3D coordinates by embedding a matrix of pairwise distance bounds between atoms. The first step is to define lower and upper bounds $d_{ij}^{\min}$, $d_{ij}^{\max}$ for each atom pair i,j , based on covalent geometry and topological distance. Then apply triangle inequality constraints to make the bounds matrix metrically consistent. Step 3 involves Sampling a distance matrix from within the bounds, and performing metric matrix embedding (eigen-decomposition) to obtain coordinates. And lastly, refine the coordinates with a force field-based minimisation.

The Experimental-Torsion Knowledge Distance Geometry (ETKDG)[8] method extends basic DG by incorporating empirical torsion-angle preferences derived from crystallographic databases and applying additional constraints for ring systems. Implemented in RDKit, ETKDG[8] produces conformers that both reproduce experimental torsion distributions and reduce the need for extensive post-embedding minimisation. For macrocycles and highly flexible molecules, ETKDG variants[21] and specialised sampling strategies are needed to explore conformational space adequately.

4.3 Energy minimisation and strain filtering

Conformers produced by OMEGA, ETKDG, or other methods are often subjected to a brief energy minimisation using a molecular mechanics force field such as MMFF94[22], UFF[23], or GAFF[24, 25]. The typical non-bonded energy expression consists of Lennard-Jones and Coulomb terms:

$$E = \sum_{i < j} \left[4\varepsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right) + \frac{q_i q_j}{4\pi\varepsilon_0\varepsilon_r r_{ij}} \right], \quad (6)$$

where i and j represent protein and ligand atoms respectively, r_{ij} is the interatomic distance, ε_{ij} and σ_{ij} are Lennard-Jones parameters, q_i q_j are partial charges, ε_0 is the vacuum permittivity and ε_r is the relative dielectric constants that accounts for screening effect of the medium.

Strain energy thresholds are applied to filter out highly strained conformers that would be unlikely in solution and could bias docking scores. In practice, these thresholds are empirical, typically 10-15 kcal/mol above the global minimum for a small molecule. However, they should be chosen carefully: overzealous filtering might weed out conformers relevant for binding, especially in tight or distorted pockets.

5 Partial charge assignment and force field parameters

5.1 Charge models

The scoring functions employed in docking typically include an electrostatic term that depends on partial atomic charges. Common small molecule charge models include an empirical charge equilibration method, such as Gasteiger-Marsili charges[26], fast and widely available in Open Babel, RD-Kit, and many docking codes. There are semiempirical-based schemes, such as AM1-BCC charges[27], which fit bond-charge corrections to reproduce HF/6-31G* electrostatic potentials. The RESP/ESP-fitted charges[28] are derived from higher-level QM calculations; common in MD force fields (GAFF/AMBER). However, these are expensive computations to handle large screening libraries. Therefore, the choice of charge model must be compatible with the docking software and, ideally, with the force field used for both the ligand

and the receptor. Some protocols (e.g., Glide/SP) use internally parameterised charges and do not expose them to the user. In contrast, others (e.g., AutoDock4, Vina) require explicit charge assignment in the input (Usually, Gasteiger charges[29]).

5.2 Parameter assignment tools

For general organic molecules, AMBER’s GAFF[24, 25] and CHARMM’s CGenFF[30] are commonly used to assign force field parameters for MD, but for docking, simpler parameterisations are often sufficient. Open-source tools such as Open Babel and RDKit can assign basic atom types and MMFF94 parameters; commercial suites provide more extensive coverage of medicinal chemistry space.

6 Software for Ligand Preparation

6.1 Open-source tools

Open Babel: Open Babel[7] is a widely used open-source cheminformatics toolkit that supports interconversion between over 100 file formats, basic geometry optimisation, charge assignment, and 3D coordinate generation. It is frequently used as a main tool in ligand preparation pipelines, especially for converting between SMILES, SDF, MOL2, and PDBQT formats.

RDKit: RDKit[16] is a modern, open-source cheminformatics library with robust functionality for molecule sanitisation, standardisation, ETKDG-based conformer generation[8], substructure search, and basic property calculations. Its Python API makes it an attractive choice for custom ligand preparation scripts.

Gypsum-DL: Gypsum-DL[10] is an open-source program specifically designed to prepare small molecule libraries for structure-based virtual screening. Given SMILES or SDF input, Gypsum-DL:

- Enumerates ionisation states (via Dimorphite-DL) over a user-defined pH range.

- Enumerates tautomers, R/S stereocenters, E/Z isomers, and ring conformers.
- Generates 3D coordinates using RDKit and filters redundant conformers.

Additional open-source utilities:

- Cinfony[31]: a Python wrapper that unifies interfaces to Open Babel, RDKit, and CDK.
- Dimorphite-DL[18]: pH-dependent protonation state predictor, often used with Gypsum-DL but can be used standalone.
- OpenEye free academic tools: limited but valuable for generating conformers and charges.

6.2 Commercial ligand preparation suites

Schrödinger’s LigPrep/Epik: LigPrep, coupled with Epik for pK_a prediction, is widely used in industry for systematic ligand preparation[3]. It performs:

- State enumeration (protonation and tautomeric) across pH.
- Stereoisomer enumeration.
- OPLS-based[32] minimisation and partial charge assignment.
- Export of docking-ready Maestro or MOL2 files.

OpenEye’s QUACPAC/OMEGA: In addition to OMEGA[9] for conformer generation, OpenEye provides QUACPAC[33] for charge state and partial charge assignment, and FILTER for removal of undesirable chemotypes. These tools are heavily used in industrial pipelines.

MOE, and Discovery Studio: Both MOE (Chemical Computing Group) and Discovery Studio (BIOVIA) implement comprehensive ligand preparation workflows (salt stripping, neutralisation, protonation, stereochemistry, conformer generation) integrated with their docking engines.

7 Verification of Ligand Structures

7.1 Chemistry and valence validation

Key checks include:

- Valence violations: Atoms with impossible valence (e.g., pentavalent carbon, pentavalent neutral nitrogen) or hypervalent halogens without appropriate hypervalent models.
- Aromaticity ring perception: Ensure aromatic rings are planar and correctly labelled; avoid inconsistent aromatic/antiaromatic assignments.
- Formal charge: Check for unexpected charge separation, zwitterions, or net charges inconsistent with known chemistry.

Automated sanitisation in RDKit or Open Babel can catch many such problems, but additional filters (e.g., PAINS, reactive groups) are commonly applied to drug-like libraries.

7.2 Geometric quality

7.2.1 Bond lengths and angles

Bond lengths and angles should be within reasonable ranges for the corresponding bond orders and atom types. Force field minimisation tends to correct these, however, conformations generated solely by distance geometry may show distorted geometries if poorly refined. Visual inspection of a subset of ligands and targeted checks for pathologies (e.g., extremely short non-bonded distances, highly bent triple bonds) are recommended.

7.2.2 Ring conformations and planarity

For aromatic and conjugated rings (e.g., benzene, heteroaromatics, amides), planarity is a useful quality criterion. Deviations from planarity can be quantified by measuring the root-mean-square deviation of ring atoms from the best-fit plane:

$$\text{RMSD}_{\text{plane}} = \sqrt{\frac{1}{N} \sum_{i=1}^N d_i^2}, \quad (7)$$

where d_i is the perpendicular distance of atom i from the best-fit plane, large deviations (e.g., $> 0.1\text{--}0.2$ Å for benzene) suggest geometric artefacts. Macrocycles and flexible rings adopt non-planar conformations, however, small aromatic rings and amide linkages should be close to planar in starting conformers.

7.3 Internal strain and unrealistic intramolecular interactions

Excessive internal strain, such as extremely close non-bonded contacts, twisted amides, and eclipsed torsions, may artificially stabilise or destabilise certain docking poses when combined with the receptor. Minimisation with a suitable force field mitigates many such issues; however, energy-based filters can be more systematic.

7.4 Comparison of prepared ligands to experimental structures

For retrospective validation and method development, it is valuable to compare prepared ligand conformations and protonation states with crystallographic reference structures from PDB, PDBbind, or curated sets (DUD-E, DEKOIS 2.0). RMSD between prepared and crystallographic conformers, as well as agreement in hydrogen-bond donor/acceptor patterns and formal charges, provide quick checks[9, 34]. Irwin and co-workers[14] have emphasised the importance of verifying that docking scores and poses correspond to chemically reasonable structures, arguing that simple metrics like RMSD (**Equation 5**) to known structures and inspection of key interactions are essential to ensure docking is *right for the right reasons*.

8 Databases and Datasets

ZINC: ZINC[14] is a free database of commercially available small molecules prepared specifically for virtual screening. It provides:

- Ready-to-dock 3D structures, often in multiple protonation and tautomeric states.
- Multiple file formats, such as SMILES, MOL2, SDF, DOCK flexibases.

- Annotated subsets, such as lead-like, fragment-like, and drug-like.

ChEMBL: ChEMBL[13] is an open, manually curated bioactivity database that curates binding, functional, and ADMET data for over a million small molecules, and thousands of protein targets. It is distributed with compound structures, typically as standardised SMILES or SDF, which can serve as input to ligand preparation pipelines. Efforts to curate stereochemistry, salt forms, and data quality for activity make it attractive for structure-based modelling. For docking, ChEMBL is primarily:

- A source of known actives and inactives for specific targets, useful for training, validation, and benchmarking.
- A source for constructing focused libraries around chemotypes of interest.

PubChem: PubChem[15], maintained by NCBI, is a large public chemical database with substance, compound, and bioassay collections, encompassing hundreds of millions of substance records and tens of millions of unique structures. Since PubChem structures originate from many depositors, a rigorous ligand preparation and standardisation step is essential before docking. It offers:

- Open access to chemical structures and associated bioactivity data.
- Powerful search tools, like substructure, similarity, and property filters.
- Links to genes, proteins, and pathways.

For docking, PubChem is often used to:

- Retrieve known ligands and analogues of a target.
- Build custom libraries around scaffolds or pharmacophores of interest.

PDB and PDBbind: The Protein Data Bank (PDB)[35] is the primary repository of macromolecular structures, including many protein–ligand complexes. These complexes define binding site geometries, water networks, and ligand poses used for docking validation and training. The PDBbind[34]

database builds on PDB by collecting experimentally measured binding affinities (K_d , K_i , and IC_{50}) for protein–ligand complexes and organising them into curated sets. PDBbind’s *refined set* is widely used to:

- Benchmark docking and scoring functions.
- Train machine learning models of binding affinity.
- Validate ligand preparation and conformer generation against crystallographic ligands.

DUD-E: DUD-E[36] is a benchmark library for evaluating docking and virtual screening methods. It allows assessment of enrichment metrics, such as ROC-AUC, early enrichment at 1%, and sensitivity to ligand preparation. Mysinger *et al.* showed that improved decoy selection, and a more diverse set of ligands in DUD-E enhance the discriminatory power of docking benchmarks. It comprises:

- 102 protein targets with known active ligands.
- Property matched decoys: compounds that resemble actives in simple physicochemical properties but are topologically dissimilar and presumed inactive.

DEKOIS 2.0: DEKOIS 2.0[6] provides challenging docking benchmark sets built from BindingDB[37] bioactivity data for 81 diverse targets. Bauer *et al.*[6, 38] used DEKOIS 2.0 to investigate the impact of ligand and protein preparation (e.g., protonation states, conformer generation) on virtual screening performance, highlighting how different preparation protocols can significantly alter enrichment. DEKOIS 2.0 emphasizes:

- Rigorous physicochemical matching of decoys to actives, including charge.
- Elimination of latent actives among decoys.

Other datasets and ML-focused docking resources: Beyond classical benchmarks, more recent datasets geared towards ML-based docking and scoring, such as Smiles2Dock[39], provide large scale collections of docked poses and scores for millions of ligand–protein pairs. These resources encourage development of docking surrogates and hybrid ML–physics-based

approaches; however, they require careful consideration of the underlying ligand preparation methods used in dataset generation.

9 Molecular Structure File Formats

9.1 SMILES

SMILES, introduced by Weininger[11, 40, 41] is a line notation for encoding molecular graphs as ASCII strings. It supports branching, ring closures, stereochemistry, and charges. SMILES is compact and human-readable, and is ubiquitous in cheminformatics databases and ligand preparation workflows. However, it does not contain 3D coordinates; conversion to 3D requires embedding using tools such as RDKit or Open Babel.

Examples:

- Ethanol: CCO or OCC
- Benzene: c1ccccc1
- Protonated amine: C[NH3+]

9.2 InChI: IUPAC International Chemical Identifier

InChI[12, 42, 43] is a non-proprietary identifier developed under IUPAC to provide a unique, canonical label for a defined chemical structure. InChI strings encode layers, such as connectivity, hydrogen, charge, stereochemistry, and isotopes, and InChIKey is a hashed, fixed-length representation suitable for indexing and web search. InChI is ideal for database linking and remove duplication, however, it is less convenient for substructure or similarity search than SMILES. It is not used directly as a docking input format, but is often used to harmonise ligand sets across databases (e.g., PDBbind–ChEMBL cross-mapping).

Example:

- InChI for ethanol is InChI=1S/C2H6O/c1-2-3/h3H,2H2,1H3
- InChIKey for ethanol is LFQSCWFLQFXTCV-UHFFFAOYSA-N
- InChI for benzene is InChI=1S/C6H6/c1-2-4-6-5-3-1/h1-6H
- InChIKey for benzene is UHOVQNZJYSORNB-UHFFFAOYSA-N

9.3 SDF/MOL: MDL connection tables

MDL MOL files (single molecules) and SDF (Structure Data File, multiple molecules) are widely used connection table formats. SDF allows arbitrary associated data fields (e.g., ID, SMILES, properties) and is a common interchange format for docking libraries. An SDF record for ethanol might look like:

```
Ethanol
  RDKit          2D

  3  2  0  0  0  0          999 V2000
    0.0000    0.0000    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
    1.5400    0.0000    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0  0  0
    2.0450    1.2300    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0  0  0
  1  2  1  0
  2  3  1  0
M  END
>  <ID>
ETHANOL

>  <SMILES>
CCO
```

9.4 MOL2: Tripos MOL2 format

MOL2 is a richer 3D format that stores atom types, charges, and substructure information. MOL2 is commonly used by docking programs such as DOCK, GOLD, and FRED, and is well supported by Open Babel and RDKit. A simplified MOL2 for ethanol might include:

```
@<TRIPOS>MOLECULE
ETHANOL
3 2 1 0 0
SMALL
NO_CHARGES

@<TRIPOS>ATOM
```

```

1 C1  0.0000  0.0000  0.0000 C.3 1 ETH 0.000
2 C2  1.5400  0.0000  0.0000 C.3 1 ETH 0.000
3 O3  2.0450  1.2300  0.0000 O.3 1 ETH 0.000

```

```
@<TRIPOS>BOND
```

```

1 1 2 1
2 2 3 1

```

9.5 PDB and PDBQT

1. PDB: The PDB format was originally developed for macromolecular structures.[35] Small-molecule ligands in PDB files are represented using HETATM records. For docking, PDB is often converted to other formats (e.g., MOL2, PDBQT) that store explicit bond orders and charges more reliably than legacy PDB conventions. A minimal ligand block:

```

HETATM  1  C1  LIG A  1  10.123  12.345  5.678  1.00 20.00  C
HETATM  2  C2  LIG A  1  11.500  12.000  5.500  1.00 20.00  C
HETATM  3  O3  LIG A  1  12.000  13.200  5.800  1.00 20.00  O

```

Column 2 shows atom numbers, column 3 is atom-type, column 4 is the residue name, column 5 shows the chain ID, column 6 is residue number, columns 7,8,9 are X,Y,Z coordinates respectively, column 10 is the occupancy, column 11 is temperature factor, and column 12 represents the elements.

2. PDBQT: PDBQT is an extension of PDB used by AutoDock and AutoDock Vina, adding partial charges (Q) and atom types (T) for the AutoDock force field. Bonds and rotatable torsions are encoded in separate sections. Conversion from SDF/MOL2 to PDBQT is typically done via AutoDockTools scripts or wrappers in Open Babel (`obabel -i sdf -o pdbqt`). For example:

```

REMARK  Autodock PDBQT file
ROOT
ATOM      1  C1  LIG      1  10.123  12.345  5.678  0.00  C
ATOM      2  C2  LIG      1  11.500  12.000  5.500  0.00  C

```

```

BRANCH 1 2
ATOM      3  O3  LIG    1   12.000   13.200   5.800  -0.50   OA
ENDBRANCH 1 2
ENDROOT
TORSDOF 1

```

9.6 Maestro (MAE)

Schrödinger’s Maestro (MAE) format and other proprietary binary/ASCII formats (e.g., SD Encrypted) store rich structural and property information. Although not human-readable, they are well supported within their respective ecosystems and often serve as the internal canonical representation during ligand preparation.

9.7 File format considerations for docking pipelines

Choice of file format for docking involves trade-offs:

- **SMILES**: compact, ideal for initial library storage and substructure search; requires 3D embedding before docking.
- **SDF**: flexible and widely supported; stores multiple molecules and arbitrary properties.
- **MOL2**: stores atom types and charges, often preferred for docking.
- **PDBQT**: required by AutoDock/Vina; less general but essential in those pipelines.

10 A Practical Ligand Preparation Workflow

This section outlines a concrete, fully open-source workflow for preparing a library of SMILES strings for docking with a generic docking engine that accepts MOL2 or PDBQT inputs.

Step 1: Input and sanitisation

1. Start from a curated SMILES library, e.g., a subset of ZINC or ChEMBL filtered for basic drug-like properties.

2. Use RDKit to parse each SMILES into a molecule object and perform sanitisation:
 - Spot valence errors and discard problematic entries.
 - Apply standardisation (functional-group normalisation, salt stripping) using predefined rules.

Step 2: Protonation and tautomer enumeration

1. For each sanitised molecule, use Gypsum-DL to enumerate protonation states and tautomers across $\text{pH } 7.4 \pm 1$:
 - Limit the number of protonation/tautomeric states per molecule (e.g., up to 16).
 - Reject highly unstable or chemically unreasonable states.
2. Alternatively, use RDKit plus an external $\text{p}K_a$ estimator or ChemAxon tools if available.

Step 3: Stereochemistry and isomer enumeration

1. Within Gypsum-DL or a custom RDKit script, enumerate undefined stereocenters and E/Z isomers subject to:
 - Restrictions on the maximum number of stereoisomers (e.g., up to 32).
 - Rules to avoid redundant enumeration of centres where chirality is irrelevant (e.g., pseudoasymmetric centres).

Step 4: 3D conformer generation

1. For each enumerated structure, run RDKit’s ETKDG to generate an initial ensemble of conformers (e.g., up to 20 per structure).
2. Optionally, supplement or validate RDKit-generated conformers with OMEGA for a subset of molecules, particularly macrocycles or highly flexible chemotypes.

Step 5: Energy minimisation and strain filtering

1. Minimise each conformer with MMFF94 via RDKit or Open Babel.
2. Compute relative strain energies and discard conformers exceeding a threshold (e.g., 10 kcal/mol above the minimum).
3. Optionally cluster conformers based on heavy-atom RMSD to ensure diversity.

Step 6: Charge assignment

1. If targeting AutoDock/Vina, assign Gasteiger partial charges using Open Babel or AutoDockTools.
2. For other docking engines, use MMFF94 or AM1-BCC charges if supported, or the engine’s internal charge model.

Step 7: Export to docking format

1. Export per-structure conformers to SDF or MOL2 using Open Babel.
2. For AutoDock/Vina, convert MOL2/SDF to PDBQT and define rotatable bonds.

Step 8: Quality Check

1. Randomly sample a few hundred ligands and visually inspect:
 - Protonation states and tautomers (do they match expected chemistry?).
 - Ring conformations and planarity of aromatics and amides.
2. Run automated checks:
 - Validate valence, bond orders, and charges.
 - Detect severe steric clashes or atypical geometries.

11 Best Practices

Robust small-molecule preparation is a prerequisite for meaningful docking-based virtual screening. Key recommendations are:

- Treat ligand preparation as a modelling step, not a trivial pre-processing task. Empirical studies show that preparation parameters can substantially affect enrichment.
- Use chemically informed protonation and tautomer states. Incorporate pK_a prediction and tautomer enumeration; avoid assuming a single arbitrary state for multi-protic systems.
- Enumerate relevant stereochemistry. Do not ignore undefined stereocenters if stereochemistry is expected to affect binding.
- Employ validated conformer generators. Use tools such as OMEGA and RDKit/ETKDG, which have been benchmarked against crystallographic data.
- Filter out chemically implausible and highly strained conformers. Apply energetic and geometric criteria to remove outliers.
- Leverage curated databases. Build libraries from ZINC, ChEMBL, PubChem, PDBbind, and established benchmarks (DUD-E, DEKOIS 2.0) to maximise data quality and facilitate comparison with literature.
- Verify. Combine automatic checks with manual inspection of representative ligands and docked poses to ensure that the modelling remains chemically reasonable.

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